Chapter 5 Lab: Two-Dimensional Plots

Instructions:

- Write a script file to do the exercises below.
- Make sure to write comments in your code and separate output display for clarity.
- Submit your script file in the following format: FirstName_LastInitial_ch5lab.m
  Ex: Ryan Cao would submit his lab as: **ryan_c_ch5lab.m**

1.

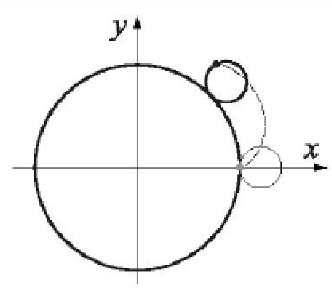
Plot the function $f(x) = x^3 - 2x^2 - 10\sin^2 x - e^{0.9x}$ and its derivative for $-2 \le x \le 4$ in one figure. Plot the function with a solid line, and the derivative with a dashed line. Add a legend and label the axes.

2.

An epicycloid is a curve (shown partly in the figure) obtained by tracing a point on a circle that rolls around a fixed circle. The parametric equation of a cycloid is given by:

$$x = 13\cos(t) - 2\cos(6.5t)$$
$$y = 13\sin(t) - 2\sin(6.5t)$$

Plot the cycloid for $0 \le t \le 4\pi$.

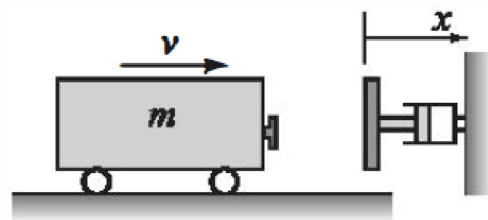


3.

Make a polar plot of the function $r = 2\sin(3\theta)\sin\theta$ for $0 \le \theta \le 2\pi$.

4.

A railroad bumper is designed to slow down a rapidly moving railroad car. After a 20,000 kg railroad car traveling at 20 m/s engages the bumper, its displacement x (in meters) and velocity v (in m/s) as a function of time t (in seconds) is given by:

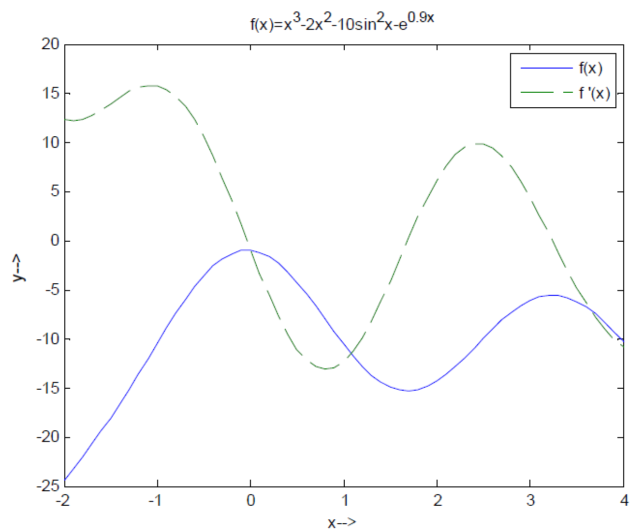


$$x(t) = 4.219(e^{-1.58t} - e^{-6.32t}) \quad \text{and} \quad v(t) = 26.67e^{-6.32t} - 6.67e^{-1.58t}$$

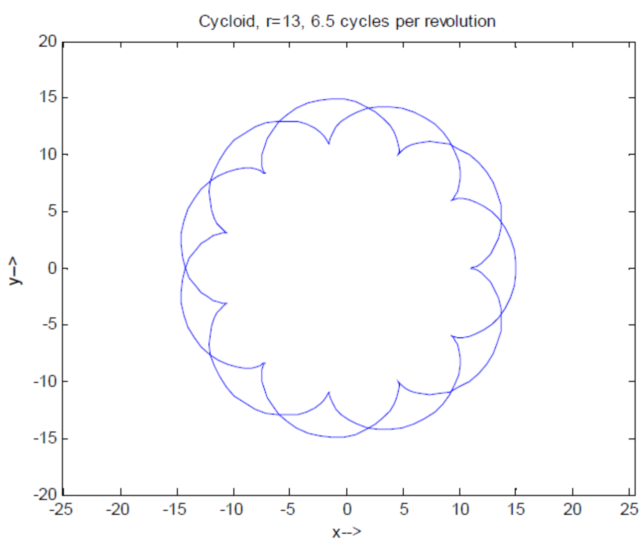
Plot the displacement and the velocity as a function of time for $0 \le t \le 4$ s. Make two plots on one page.

Output:

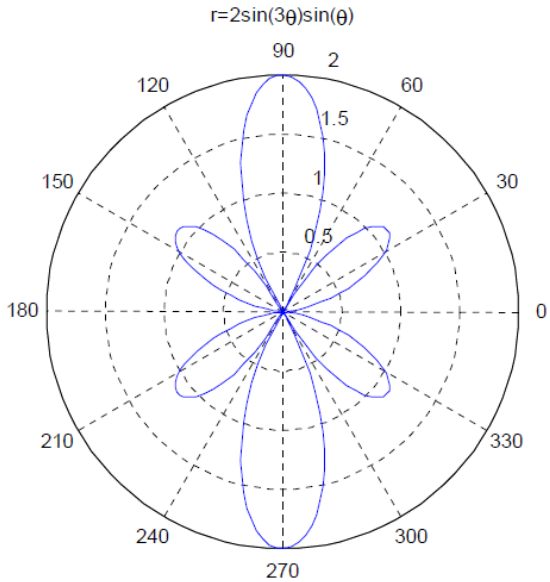
Problem 1:



Problem 2:



Problem 3:



Problem 4: